United States Patent	12415302
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Yoshida; Syuhei

Conveying apparatus, resin molding apparatus, conveying method, and resin molded product manufacturing method

Abstract

The present invention is a transfer device for unloading a resin molded article from one mold provided with a resin injection unit, the resin molded article being unloaded by the transfer device from one mold provided with a resin injection unit having a member projecting between at least part of the resin molded article and another mold, wherein the resin injection unit has a member projecting between at least part of the resin molded article and the other mold, and the transfer device is equipped with a holding member for holding the resin molded article and a horizontal movement mechanism that causes the holding member holding the resin molded article to move in the horizontal direction to separate the resin molded article from the resin injection unit.

Inventors:	**Yoshida; Syuhei (Kyoto, JP)**
Applicant:	**TOWA CORPORATION** (Kyoto, JP)
Family ID:	**1000008818359**
Assignee:	**TOWA CORPORATION (N/A, JP)**
Appl. No.:	**17/286653**
Filed (or PCT Filed):	**August 01, 2019**
PCT No.:	**PCT/JP2019/030369**
PCT Pub. No.:	**WO2020/079923**
PCT Pub. Date:	April 23, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210370565 A1	Dec. 02, 2021

Foreign Application Priority Data

JP	2018-197518	Oct. 19, 2018

Publication Classification

Int. Cl.: **B29C45/00** (20060101); **B21D39/00** (20060101); **B21D39/20** (20060101); **B29C37/00** (20060101); **B29C43/00** (20060101); **B29C43/02** (20060101); **B29C43/50** (20060101); **B29C44/00** (20060101); **B29C44/20** (20060101); **B29C45/02** (20060101); **B29C45/42** (20060101); B29L31/34 (20060101)

U.S. Cl.:

CPC **B29C45/4225** (20130101); **B29C43/02** (20130101); **B29C43/50** (20130101); **B29C45/02** (20130101); B29C2043/5046 (20130101); B29L2031/3425 (20130101)

Field of Classification Search

CPC: B29C (43/50); B29C (45/4225); B29C (45/02)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5603879	12/1996	Osada	264/102	B29C 45/34
5753538	12/1997	Kuno	264/272.17	B29C 45/14655
5846477	12/1997	Hotta	264/272.17	B29C 45/14655
5885506	12/1998	Bolanos	264/328.5	B29C 45/463
6048483	12/1999	Miyajima	264/272.17	H01L 21/566
6439869	12/2001	Seng	264/272.17	B29C 45/14655
2002/0056942	12/2001	Seng	264/272.17	B29C 45/02
2004/0063234	12/2003	Takase	438/26	H01L 24/97
2006/0131780	12/2005	Kuno	425/444	B29C 41/42
2009/0304843	12/2008	Steiner	425/595	B29C 45/0003
2012/0175812	12/2011	Ikeda	425/129.1	H01L 21/565
2016/0001901	12/2015	Takamatsu	53/111R	B29C 45/7626
2018/0117813	12/2017	Ho	N/A	B29C 45/1701
2021/0107186	12/2020	Endoh	N/A	B29C 51/42
2021/0387385	12/2020	Yoshida	N/A	H01L 21/56
2022/0161475	12/2021	Okunishi	N/A	B29C 45/56
2024/0009900	12/2023	Yoshida	N/A	B29C 45/2602

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104112679	12/2013	CN	N/A
4541547	12/2024	EP	B29C 33/44
08-197567	12/1995	JP	N/A
H10335360	12/1996	JP	H01L 21/565
10-335360	12/1997	JP	N/A
H10335360	12/1997	JP	N/A
H-11284002	12/1998	JP	B29C 45/38
2001-217269	12/2000	JP	N/A
2004-193582	12/2003	JP	N/A
2014-212246	12/2013	JP	N/A
2014212246	12/2013	JP	N/A
2017121771	12/2016	JP	N/A
2022039709	12/2021	JP	B29C 33/44
20120028639	12/2011	KR	N/A
201234502	12/2011	TW	N/A
WO-2018180409	12/2017	WO	N/A

OTHER PUBLICATIONS

JPH10335360A (Kishi) May 1997 (online machine translation), [Retrieved on Jun. 8, 2023]. Retrieved from: Espacenet (Year: 1997). cited by examiner
JPH10335360 (Yuichi) Dec. 1998 (online machine translation), [Retrieved on Aug. 7, 2023]. Retrieved from: Espacenet (Year: 1998). cited by examiner
JP2014212246 (Shinj) Nov. 2014 (online machine translation), [Retrieved on Aug. 7, 2023]. Retrieved from: Espacenet (Year: 2014). cited by examiner
WO-2018180409 (Katsuhisa) Oct. 4, 2018 (online machine translation), [Retrieved on Aug. 7, 2023]. Retrieved from: Espacenet (Year: 2018). cited by examiner
JPH-11284002-A (Nakayama) Oct. 1999 (online machine translation), [Retrieved on May 22, 2025]. Retrieved from: Espacenet (Year :1999). cited by examiner
JP-2022039709-A (Yoshida) Mar. 2022 (online machine translation), [Retrieved on May 22, 2025]. Retrieved from: Espacenet (Year: 2022). cited by examiner
Office Action dated May 11, 2022 issued in CN patent application No. 201980068065.0, 12 pgs. cited by applicant
Office Action dated Sep. 3, 2019 issued for Japanese patent application No. 2018-197518, 12 pgs. cited by applicant
International Search Report dated Sep. 10, 2019 issued for International application No. PCT/JP2019/030369, 4 pgs. cited by applicant
Decision to grant a patent dated Jan. 9, 2020 issued for Japanese patent application No. 2018-197518t, 6 pgs. cited by applicant
Office Action dated Jan. 9, 2020 issued for Taiwan patent application No. 108132124, 9 pgs. cited by applicant
Decision of refusal dated Jun. 16, 2020 issued for Taiwan patent application No. 108132124, 5 pgs. cited by applicant

---

Primary Examiner: Hindenlang; Alison L

Assistant Examiner: Behrens, Jr.; Andres E.

Attorney, Agent or Firm: Brooks Kushman P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. national phase of PCT Application No. PCT/JP2019/030369, filed Aug. 1, 2019, which claims priority to Japanese Patent Application No. 2018-197518, filed Oct. 19, 2018, which are both incorporated by reference herein in their entireties.

TECHNICAL FIELD

(2) The present invention relates to a conveying apparatus, a resin molding apparatus, a conveying method, and a resin molded product manufacturing method.

TECHNICAL BACKGROUND

(3) Conventionally, as is shown, for example, in Patent Document 1, in a resin molding apparatus that includes a lower mold in which is formed a pot into which a resin material is placed, and an upper mold in which is formed a cavity into which the resin material placed in the pot is injected, a structure is employed in which a resin molded product is extracted by a conveying apparatus.

DOCUMENTS OF THE PRIOR ART

Patent Documents

(4) [Patent Document 1] Japanese Unexamined Patent Application Laid-Open (JP-A) No. 2001-217269

DISCLOSURE OF THE INVENTION

Problems to be Solved by the Invention

(5) However, in a conventional method in which a substrate that has been resin molded using this resin molding apparatus is lifted up from the lower mold and conveyed away, if, for example, a component from the lower mold is present between the substrate and the upper mold, then this component ends up coming into contact with the substrate and it may become difficult for the substrate to be conveyed away.

(6) The present invention was conceived in order to solve the above-described drawbacks, and it is a principal object thereof to enable a resin molded product to be conveyed away by a conveying apparatus from one mold in which is provided a component that protrudes between at least a portion of the resin molded product and another mold.

Means for Solving the Problem

(7) In other words, a conveying apparatus according to the present invention is a conveying apparatus that conveys a resin molded product away from one mold in which a resin injection portion is provided, and is characterized in that the resin injection portion is provided with a component that protrudes between at least a portion of the resin molded product and another mold, and in that the conveying apparatus is provided with a holding component that holds the resin molded product, and a horizontal movement mechanism that moves the holding component which is holding the resin molded product in a horizontal direction so as to move the holding component away from the resin injection portion.

Effects of the Invention

(8) According to the present invention which is formed in this manner, a resin molded product is able to be conveyed away by a conveying apparatus from one mold in which is provided a resin injection portion which has a component that protrudes between at least a portion of the resin molded product and another mold.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic view showing a structure of a resin molding apparatus according to an embodiment of the present invention.

(2) FIG. **2** is a schematic view showing a substrate placement state of a molding module of the same embodiment.

(3) FIG. **3** is a schematic view showing a resin material loaded state of the molding module of the same embodiment.

(4) FIG. **4** is a schematic view showing a mold fastened state of the molding module of the same embodiment.

(5) FIG. **5** is a schematic view showing a resin injected state of the molding module of the same embodiment.

(6) FIG. **6** is a schematic view showing a mold opened state of the molding module of the same embodiment.

(7) FIG. **7** is a side view schematically showing a state in which the conveying apparatus of the same embodiment is located above a lower mold.

(8) FIG. **8** contains views schematically showing structures of a horizontal movement mechanism and a vertical movement mechanism of the same embodiment.

(9) FIG. **9** contains views schematically showing operations of the horizontal movement mechanism and the vertical movement mechanism of the same embodiment.

(10) FIG. **10** is a side view schematically showing a state in which the conveying apparatus of the same embodiment is in a predetermined holding position.

(11) FIG. **11** is a side view schematically showing a state in which the conveying apparatus of the same embodiment is holding a resin molded product using suction.

(12) FIG. **12** is a side view schematically showing a state in which the conveying apparatus of the same embodiment is moving a resin molded product using the horizontal movement mechanism.

(13) FIG. **13** is a side view schematically showing a state in which the conveying apparatus of the same embodiment is lifting up a resin molded product using the vertical movement mechanism.

(14) FIG. **14** is a side view schematically showing a state in which a distance between conveying claws has been shortened in the conveying apparatus of the same embodiment.

(15) FIG. **15** is a schematic view showing a positional relationship between a resin molded product and a cull block.

DESCRIPTION OF THE REFERENCE NUMERALS

(16) **100** . . . Resin Molding Apparatus W**2** . . . Resin Molded Product **14** . . . Resin Injection Portion **141***b* . . . Protruding Portion **141** . . . Cull Block **19** . . . Unloader (Conveying Apparatus) **20** . . . Holding Component **22** . . . Horizontal Movement Mechanism **23** . . . Vertical Movement Mechanism **21** . . . Base Component **221** . . . Guide Portion **222** . . . Horizontal Drive Portion **231** . . . Guide Portion **232** . . . Vertical Cam Mechanism **222***a* . . . Slide Component **223** . . . Link Mechanism **224** . . . First Horizontal Cam Mechanism **225** . . . Second Horizontal Cam Mechanism **24** . . . Suction Pads (Mold Release Mechanism) **25** . . . Conveying Claws

Best Embodiments for Implementing the Invention

(17) Next, the present invention will be described in further detail using examples. It should be noted, however, that the present invention is not limited by the following description.

(18) As is shown in FIG. **15**, in a resin molding apparatus, a cull block in which is formed a pot that contains a resin material may be provided in a lower mold. This cull block protrudes upwards from an upper surface of the lower mold. The cull block also has a protruding portion that protrudes into an upper portion of a substrate that has been placed on the upper surface of the lower mold. In a state in which pot-side end portions of the substrate are sandwiched between the upper surface of the lower mold and a lower surface of the protruding portion, the mold is clamped together. When the mold is in this clamped state, a resin flow path is formed between the upper portion and

protruding portion of the cull block and a recessed portion in the upper mold, and resin material is injected into a cavity through this resin flow path. In this resin molding apparatus, the resin material is injected from the upper surface of the substrate by means of this resin flow path.
(19) However, in a case in which a substrate that has been resin molded using this resin molding apparatus is to be lifted up from the lower mold and conveyed away, the substrate comes into contact with the protruding portion of the cull block so that it becomes difficult to perform the conveying-away operation.
(20) As is described above, the conveying apparatus according to the present invention is a conveying apparatus that conveys a resin molded product away from one mold in which a resin injection portion is provided, and is characterized in that the resin injection portion is provided with a component that protrudes between at least a portion of the resin molded product and another mold, and in that the conveying apparatus is provided with a holding component that holds the resin molded product, and a horizontal movement mechanism that moves the holding component which is holding the resin molded product in a horizontal direction so as to move the holding component away from the resin injection portion.
(21) By employing this type of conveying apparatus, because the conveying apparatus includes the horizontal movement mechanism that moves the holding component which is holding the resin molded product in a horizontal direction so as to move the holding component away from the resin injection portion, it is possible, using this conveying apparatus, to convey a resin molded product away from one mold in which is provided a resin injection portion which has a component that protrudes between at least a portion of the resin molded product and another mold.
(22) Moreover, it is desirable that the conveying apparatus be further provided with a vertical movement mechanism that moves the holding component which has been moved by the horizontal movement mechanism in a direction away from the one mold and towards the other mold.
(23) Additionally providing the conveying apparatus with a base component in which the holding component is movably provided may also be considered. As a specific embodiment of the horizontal movement mechanism in this structure, providing the horizontal movement mechanism with a guide portion that guides the holding component away from the resin injection portion, and a horizontal drive portion that causes the holding component to move along the guide portion may also be considered. In addition, as a specific embodiment of the vertical movement mechanism, providing the vertical movement mechanism with a guide portion that guides the holding component in a vertical direction, and a vertical cam mechanism that enables the holding component to move in a vertical direction in conjunction with a horizontal movement of the holding component performed by the horizontal drive portion may also be considered.
(24) If this structure is employed, because the same drive portion can be used to drive both the horizontal movement mechanism and the vertical movement mechanism, not only can a simplification of the structure be achieved, but a reduction in costs also becomes possible.
(25) It is also desirable that the horizontal drive portion include a slide component that slides in a direction towards or in a direction away from the resin injection portion, and that the horizontal movement mechanism be further provided with a link mechanism that enables the holding component to move in a direction away from the resin injection portion in conjunction with a movement of the slide component.
(26) If this structure is employed, because the horizontal movement mechanism includes a link mechanism, a movement of the slide component can be transmitted without any difficulty to a movement of the holding component.
(27) In order to interlink the slide component, the link mechanism, and the holding component irrespectively of their individual movement directions, it is desirable that a first horizontal cam mechanism be provided between the slide component and the link mechanism, and that a second horizontal cam mechanism be provided between the link mechanism and the holding component.
(28) It is also desirable that the conveying apparatus be further provided with a mold release

mechanism that releases the resin molded product from the one mold, and that the horizontal movement mechanism move the holding component in a horizontal direction in a state in which the resin molded product has been released from the one mold by the mold release mechanism.
(29) If this structure is employed, then in a case in which the resin molded product is to be moved, because the resin molded product is moved after having been released from the one mold, it is possible to prevent the resin molded product from becoming damaged or broken. In addition, the moving of the resin molded product can be performed smoothly.
(30) In order to hold the resin molded product, providing the holding component with suction pads that suction the resin molded product may also be considered. In this structure, in order to both simplify the structure of the conveying mechanism and additionally enable a reduction in costs to be achieved, it is desirable that the mold release mechanism be formed using suction force from the suction pads.
(31) Forming the conveying apparatus such that the conveying apparatus is additionally provided with conveying claws that hold the resin molded product by grasping it might also be considered. Because using these conveying claws enables a resin molded product to be conveyed more stably, it is possible to inhibit the resin molded product from being dropped.
(32) In this structure, it is desirable that, after the holding component has been moved in a direction away from the one mold towards the other mold using the vertical movement mechanism, the resin molded product be held using the conveying claws.
(33) In a structure that includes these conveying claws, by firstly moving the resin molded product horizontally and then moving it vertically, the resin molded product can be released from the one mold, and a surface on the one mold side of the resin molded product can be held using the conveying claws.
(34) Moreover, by additionally providing the conveying apparatus with a vertical movement mechanism, it becomes possible, without having to provide a releasing component (such as, for example, an ejector pin) in one mold in order for the resin molded product to be released from that one mold, to hold a surface on the one mold side of the resin molded product using the conveying claws.
(35) Moreover, a resin molding apparatus according to the present invention is characterized in including the above-described conveying apparatus.
(36) If this resin molding apparatus is employed, then it is possible, using a conveying apparatus, to convey a resin molded product away from one mold in which is provided a resin injection portion which has a component that protrudes between at least a portion of the resin molded product and another mold.
(37) Furthermore, a conveying method for a resin molded product of the present invention is characterized in that, using the above-described conveying apparatus, a holding component that is holding a resin molded product which has been resin sealed is moved in a horizontal direction by the horizontal movement mechanism so as to move away from the resin injection portion, and the holding component that has been moved away from the resin injection portion is then moved by the vertical movement mechanism in the direction of the other mold and is conveyed out.
(38) In addition to this, a resin molded product manufacturing method according to the present invention is a resin molded product manufacturing method in which an electronic component is resin molded by means of resin molding, and that is provided with a molding step in which resin molding is performed on an object to be molded, and a conveying away step in which a resin molded product that has been resin molded is conveyed away using the above-described conveying apparatus.
(39) If this resin molded product manufacturing method is employed, then it is possible, using a conveying apparatus, to convey a resin molded product away from one mold in which is provided a resin injection portion which includes a component that protrudes between at least a portion of the resin molded product and another mold.

Embodiment of the Present Invention

(40) Hereinafter, an embodiment of a resin molding apparatus according to the present invention will be described with reference to the drawings. Note that each of the drawings shown below has been drawn schematically with omissions or enhancements being employed where appropriate in order to make the drawings easier to understand. In addition, component elements that are mutually the same are given the same descriptive symbols and any duplicated description thereof is omitted.

Overall Structure of a Resin Molding Apparatus

(41) A resin molding apparatus **100** of the present embodiment performs resin molding on an object being molded W**1** on which electronic components Wx have been packaged by performing transfer molding thereon using a resin material J.

(42) Here, the object being molded W**1** may be, for example, a metal substrate, a resin substrate, a glass substrate, a ceramic substrate, a circuit substrate, a semiconductor substrate, a wiring substrate, or a lead frame or the like, and may either include or not include wiring. The resin material J used for the resin molding may, for example, be a composite material containing a thermosetting resin, and the resin material J may be in granular form, powder form, liquid form, sheet form, or in tablet form or the like. Examples of the electronic components Wx packaged on an upper surface of the object being molded W**1** include bare chips and resin sealed chips.

(43) More specifically, as is shown in FIG. **1**, the resin molding apparatus **100** is provided with various component elements including a supply module **100**A that supplies objects being molded W**1** before molding and also the resin material J, a molding module **100**B that performs the resin molding, and a storage module **100**C that stores objects being molded W**2** after the molding (hereinafter, these are referred to as resin molded products W**2**). Note that each of the supply module **100**A, the molding module **100**B, and the storage module **100**C is able to be combined with or separated from the other component elements, and is also replaceable.

(44) The supply module **100**A is provided with an object being molded supply portion **11** that supplies the objects being molded W**1**, a resin material supply portion **12** that supplies the resin material J, and a conveying apparatus **13** (hereinafter, referred to as a loader **13**) that receives the objects being molded W**1** from the object being molded supply portion **11** and conveys them to the molding module **100**B, and also receives the resin material J from the resin material supply portion **12** and conveys it to the molding module **100**B.

(45) The loader **13** moves back and forth between the supply module **100**A and the molding module **100**B, and performs this by moving along rails (not shown in the drawings) which are provided between the supply module **100**A and the molding module **100**B.

(46) As is shown in FIG. **2** through FIG. **6**, the molding module **100**B has one mold **15** (hereinafter, referred to as a lower mold **15**) of a forming mold in which a resin injection portion **14** is provided, another mold **16** (hereinafter, referred to as an upper mold **16**) of the forming mold which is provided so as to face the lower mold **15** and in which is formed a cavity **16***a* into which the resin material J is injected, and a mold clamping mechanism **17** that clamps the lower mold **15** and the upper mold **16** together. The lower mold **15** is provided via a lower platen **102** on a movable plate **101** that is raised and lowered by the mold clamping mechanism **17**. The upper mold **16** is provided via an upper platen **103** in an upper fixed plate (not shown in the drawings).

(47) The resin injection portion **14** of the present embodiment has a cull block **141** in which is formed a pot **141***a* that contains the resin material J, a plunger **142** that pumps out the resin material J contained in the pot **141***a*, and a plunger drive portion **143** that drives the plunger **142**. Note that the plunger **142** pumps out the resin material J after this has been heated in the pot **141***a* so as to become molten.

(48) The cull block **141** is provided in a recessed portion **15**M formed in the lower mold **15**, and is elastically supported by an elastic component **144** so as to be able to move up and down inside the recessed portion **15**M. In addition, a protruding portion **141***b* that protrudes outwards beyond an aperture of the recessed portion **15**M is formed in an upper end portion of the cull block **141**. The

cull block **141** that includes the protruding portion **141***b* in this way acts as a component that protrudes between at least a portion of the resin molded product W**2** and the upper mold **16**.

(49) A cavity **16***a* that contains the electronic components Wx of the object being molded W**1** and into which the molten resin material J is injected is formed in the upper mold **16**. In addition, a cull **16***b*, which is a recessed portion, is formed in a portion of the upper mold **16** that faces the protruding portion **141***b* of the cull block **141**, and a runner **16***c* that connects the cull **16***b* to the cavity **16***a* is also formed in the upper mold **16**. Note that, although not shown in the drawings, an air vent is formed in the upper mold **16** on the opposite side from the cull block **141**.

(50) Moreover, as is shown in FIG. **4**, when the lower mold **15** and the upper mold **16** are clamped together by the mold clamping mechanism **17**, a resin flow path formed by the cull **16***b* and the runner **16***c* connects the pot **141***a* to the cavity **16***a*. In addition, when the lower mold **15** and the upper mold **16** are clamped together, a pot-side end portion of the object being molded W**1** is sandwiched between a lower surface of the protruding portion **141***b* of the cull block **141** and an upper surface of the lower mold **15**. In this state, if the molten resin material J is injected into the cavity **16***a* by the plunger **142**, then the electronic components Wx of the object being molded W**1** are resin sealed.

(51) The storage module **100**C is provided with a storage portion **18** that stores the resin molded products W**2**, and a conveying apparatus **19** (hereinafter, referred to as an unloader **19**) that receives the resin molded products W**2** from the molding module **100**B and conveys them to the storage portion **18**.

(52) The unloader **19** moves back and forth between the molding module **100**B and the storage module **100**C, and performs this by moving along rails (not shown in the drawings) provided between the molding module **100**B and the storage module **100**C. Note that the unloader **19** is described below in detail.

(53) A simple description of basic operations of this resin molding apparatus **100** will now be given with reference to FIG. **2** through FIG. **6**.

(54) As is shown in FIG. **2**, in a state in which the lower mold **15** and the upper mold **16** are open, an object being molded W**1** before molding is conveyed by the loader **13**, and is then transferred to the lower mold **15** and placed thereon. At this time, the upper mold **16** and the lower mold **15** have been heated to a temperature that enables the resin material J to be melted and cured. Thereafter, as is shown in FIG. **3**, the resin material J is conveyed by the loader **13** and is then held within the pot **141***a* of the cull block **141**.

(55) If, in this state, the lower mold **15** is raised by the mold clamping mechanism **17**, then as is shown in FIG. **4**, the cull block **141** hits against the upper mold **16** and is lowered down within the recessed portion **15**M of the lower mold **15**, so that the lower surface of the protruding portion **141***b* comes into contact with the pot-side end portion of the object being molded W**1**. In addition, the lower surface of the upper mold **16** comes into contact with the vent-side end portion of the object being molded W**1**. As a result, the lower mold **15** and the upper mold **16** are clamped together. Once they are clamped together, if the plunger **142** is then raised by the plunger drive portion **143**, then as is shown in FIG. **5**, the molten resin material J inside the pot **141***a* passes through the resin flow path and is injected into the cavity **16***a*. After the resin material J has subsequently cured inside the cavity **16***a*, then as is shown in FIG. **6**, the mold is opened by the mold fastening mechanism **17**, and the resin molded product W**2** is conveyed away by the unloader **19**, and is transported to the storage portion **18**.

Specific Structure of the Unloader **19**

(56) Next, the specific structure of the unloader **19** of the present embodiment will be described with reference to FIG. **7** through FIG. **13**. Note that, in FIG. **7**, FIG. **10**, and FIG. **11**, conveying claws **25** have been omitted from the drawings.

(57) As is shown in FIG. **7** and the like, the unloader **19** is provided with a holding component **20** that holds the resin molded products W**2**, a base component **21** on which the holding component **21**

is movably provided, a horizontal movement mechanism **22** that moves the holding component **20** which is holding the resin molded product W**2** in a horizontal direction away from the cull block **141**, and a vertical movement mechanism **23** that moves the holding component **20**, which has been moved by the horizontal movement mechanism **22**, in an upward direction from the lower mold **15** towards the upper mold **16**.
(58) The holding component **20** includes suction pads **24** that suction a resin molded product W**2** that has been placed on the lower mold **15**. The suction pads **24** are provided on a lower surface of the holding component **20**, and a suction mechanism (not shown in the drawings) that is used to suction the resin molded product W**2** is connected to the suction pads **24**. Note that the suction mechanism includes, for example, a suction flow path that is formed within the holding component, and a suction pump that is connected to the suction flow path.
(59) The suction pads **24** of the present embodiment form part of a mold release mechanism that releases the resin molded product W**2** from the lower mold **15**. When a resin molded product W**2** is suctioned by the suction pads **24**, the suction pads **24** are elastically deformed by the suction force from the suction pads **24** and the resin molded product W**2** is lifted up from the lower mold **15** (see FIG. **11**). Note that the amount by which the resin molded product W**2** is lifted up by the suction pads **24** is not so great as to cause it to come into contact with the protruding portion **141***b* of the cull block **141**.
(60) The base component **21** is provided so as to be able to move along rails (not shown in the drawings) in a horizontal direction and in a vertical direction. Conveying claws **25** (see FIG. **12**-FIG. **14**) that are used to prevent a resin molded product W**2** that is being held by the holding component **20** from being dropped, or are used to hold a resin molded product W**2** are provided on the base component **21**. A pair of these conveying claws **25** are provided so as to grasp both end portions of a resin molded product W**2**, and a distance between the conveying claws **25** is widened or shortened by a drive portion (not shown in the drawings). The resin molded product W**2** is able to be grasped as a result of the drive portion shortening the distance between the conveying claws **25**, and as a result of this distance between the conveying claws **25** being widened, the resin molded product W**2** is able to be moved upwards or downwards by the vertical movement mechanism **23** without being obstructed by the conveying claws **25**.
(61) The horizontal movement mechanism **22** moves the holding component **20** which is holding a resin molded product W**2** rectilinearly in a direction away from the cull block **141**. Note that, as is described above, because the resin molded product W**2** is released from the mold by the suction pads **24** of the holding component **20**, the horizontal movement mechanism **22** causes a resin molded product W**2** released from the mold by the suction pads **24** to move rectilinearly in a direction away from the cull block **141**.
(62) More specifically, the horizontal movement mechanism **22** includes a guide portion **221** that guides the holding component **20** in a direction away from the cull block **141**, and a horizontal drive portion **222** that moves the holding portion **20** along the guide portion **221**.
(63) The guide portion **221** includes rails that are provided extending in a front-rear direction on the base component **21** towards and away from the cull block **141**, and a slider that is provided on the holding component **20** and moves along the rails.
(64) As is shown in FIG. **8**, the horizontal drive portion **222** includes a slide component **222***a* that slides either in a direction towards or in a direction away from the cull block **141** (i.e., the pot side), and is able to move the slide component **222***a* using an actuator **222***b* such as, for example, an air cylinder.
(65) Moreover, as is shown in FIG. **8**, the horizontal movement mechanism **22** is additionally provided with a link mechanism **223** that enables the holding component **20** to be moved in a direction away from the cull block **141** in conjunction with a movement of the slide component **222***a*. The link mechanism **223** of the present embodiment is formed so as to enable the holding component **20** to move in a direction away from the cull block **141** in conjunction with the slide

component **222***a* moving in a direction away from the cull block **141**.

(66) More specifically, the link mechanism **223** includes two link arms **223***a* and **223***b*, and the first link arm **223***a* is provided closer to the slide component **222***a* side than the second link arm **223***b*. Each of the link arms **223***a* and **223***b* is able to rotate around rotation shafts **223***a***1** and **223***b***1** that are provided on the base component **21**. An interlink portion **223***c* between the first link arm **223***a* and the second link arm **223***b* is formed by a slide hole **223***c***1** that is provided in the one link arm **223***b*, and a pin **223***c***2** that is provided on the other link arm **223***a* and slides inside the slide hole **223***c***1**. As a result, in conjunction with a rotation of the first link arm **223***a*, the second link arm **223***b* is rotated in an opposite direction from the rotation direction of the first link arm **223***a*.

(67) Furthermore, as is shown in FIG. **8**, in the horizontal movement mechanism **22**, a first horizontal cam mechanism **224** is provided between the slide component **222***a* and the link mechanism **223**, and a second horizontal cam mechanism **225** is provided between the link mechanism **223** and the holding component **20**.

(68) The first horizontal cam mechanism **224** includes a first roller contact portion **224***a* that is provided in the slide component **222***a*, and a first roller component **224***b* that is provided on the first link arm **223***a* and that moves in conjunction with a movement of the first roller contact portion **224***a*. The first roller contact portion **224***a* of the present embodiment is formed by an inclined surface and a flat surface that are formed on a side surface of the slide component **222***a*, and is formed such that the first roller component **224***b* slides over this inclined surface and flat surface. As a result of the first roller component **224***b* moving over the inclined surface of the first roller contact portion **224***a*, the first link arm **223***a* is rotated in one direction (see FIG. **9** (*b*)). In contrast, as a result of the first roller component **224***b* moving over the flat surface of the first roller contact portion **224***a*, the first link arm **223***a* is maintained in its existing state without being rotated (see FIG. **9** (*c*)). The first roller component **224***b* of the present embodiment may also be formed, for example, using rolling bodies (i.e., bearings) and bushes that are rotatably provided on an inner side end portion of the first link arm **223***a*. As a result, the movement of the first horizontal cam mechanism **224** (i.e., the rotation of the roller component **224***b*) is able to proceed smoothly.

(69) The second horizontal cam mechanism **225** includes a second roller contact portion **225***a* that is provided in the holding component **20**, and a second roller component **225***b* that is provided on the second link arm **223***b* and that moves in conjunction with a movement of the second roller contact portion **225***a*. The second roller component **225***b* of the present embodiment may also be formed, for example, using rolling bodies (i.e., bearings) and bushes that are rotatably provided on an outer side end portion of the second link arm **223***b*. As a result, the movement of the second horizontal cam mechanism **225** (i.e., the rotation of the roller component **225***b*) is able to proceed smoothly. In addition, the second roller contact portion **225***a* is provided on a surface of the holding component **20** that faces toward the cull block **141** side, and is formed as a flat surface in the present embodiment.

(70) By employing this type of link mechanism **223**, if the horizontal drive portion **222** moves the slide component **222***a* in a direction away from the cull block **141** (i.e., from FIG. **9** (*a*) to FIG. **9** (*b*)), then the first roller component **224***b* moves along the inclined surface of the first roller contact portion **224***a* of the slide component **222***a*, and the first link arm **223***a* is rotated. In conjunction with this rotation of the first link arm **223***a*, the second link arm **223***b* is rotated in the opposite direction from the rotation direction of the first link arm **223***a*, and the second roller component **225***b* of the second link arm **223***b* pushes the second roller contact portion **225***a* in a direction away from the cull block **141**. As a result of this, the holding component **20** is moved in a direction away from the cull block **141**. Thereafter, when the first roller component **224***b* reaches the flat surface of the first roller contact portion **224***a* (i.e., from FIG. **9** (*b*) to FIG. **9** (*c*)), the rotation of the first link arm **223***a* is stopped, and the movement of the holding component **20** in a horizontal direction is also stopped.

(71) Note that the holding component **20** is urged towards the cull block **141** side, for example, by

an elastic component **26** that is provided via a T-shaped block **226**, and when the slide component **222***a* is moved by the horizontal drive portion **222** towards the cull block **141**, the holding portion **20** is also moved towards the cull block **141**.

(72) The vertical movement mechanism **23** enables the holding component **20** which has been moved by the horizontal movement mechanism **22** to be moved rectilinearly in a vertical direction. Note that when this movement is being performed by the vertical movement mechanism, the resin molded product W**2** is positioned on an outer side of the protruding portion **141***b* of the cull block **141**.

(73) More specifically, the vertical movement mechanism **23** includes a guide portion **231** (see FIG. **7**) that guides the holding component **20** in a vertical direction (i.e., in an up-down direction), and a vertical cam mechanism **232** (see FIG. **8**) that enables the holding component **20** to be moved in a vertical direction in conjunction with the horizontal movement of the holding component **20** that is performed by the horizontal drive portion **222**.

(74) The guide portion **231** includes rails that are provided extending in an up-down direction on the base component **21**, and a slider that is provided on the holding component **20** and that moves along the rails.

(75) As is shown in FIG. **8**, the vertical cam mechanism **232** includes a vertical movement roller component **232***a* that is provided in the slide component **222***a*, and a vertical movement roller contact portion **232***b* that is provided in the holding component **20** and is moved in conjunction with a movement of the vertical movement roller component **232***a*. The vertical movement roller component **232***a* of the present embodiment moves together with the slide component **222***a*. In addition, the vertical movement roller contact portion **232***b* has an inclined surface that causes the vertical movement roller contact portion **232***b* to move in an upward direction in conjunction with the movement of the vertical movement roller component **232***a* in a direction way from the cull block **141**.

(76) When the horizontal drive portion **222** moves the slide component **222***a* in a direction away from the cull block **141** using this vertical movement mechanism **23** (i.e., from FIG. **9** (*b*) to FIG. **9** (*c*)), the vertical movement roller component **232***a* moves in a direction away from the cull block **141**, and the vertical movement roller contact portion **232***b* moves in an upward direction (see FIG. **9** (*c*)). As a result, the holding component **20** is moved in an upward direction.

(77) Note that, when the slide component **222***a* is moved towards the cull block **141** by the horizontal drive portion **222**, the holding component **20** is moved by its own weight in a downward direction along the guide portion **231** (see FIG. **9** (*b*)).

Operations of the Unloader **19**

(78) Next, conveying operations performed by the unloader **19** will be described with reference to FIG. **7**, and FIG. **10** through FIG. **14**.

(79) Firstly, as is shown in FIG. **7**, after the unloader **19** has been moved to above the lower mold **15**, as is shown in FIG. **10**, the resin molded product W**2** is lowered to a predetermined position in order to be held via suction. Next, as is shown in FIG. **11**, the resin molded product W**2** is held by suction provided by the suction pads **24** of the holding component **20**. Note that, although not shown in FIG. **11**, the distance between the conveying claws **25** has been widened in order that the conveying claws **25** do not obstruct the holding via suction of the resin molded product W**2**. In addition, a suction pad **27** that is used to suction excess resin K remaining on the upper surface of the cull block **141** is provided in the base component **21**, and this suction pad **27** suctions and holds excess resin K. Here, as is shown in FIG. **11**, the resin molded product W**2** is released from the lower mold **15** simultaneously with the resin molded product W**2** being held via suction by the suction pads **24** of the holding component **20**.

(80) Next, as is shown in FIG. **12**, the holding component **20** is moved by the horizontal movement mechanism **22** in a direction away from the cull block **141**. More specifically, when slide component **222***a* is moved in an away direction by the horizontal drive portion **222** (see FIG. **9** (*b*)),

the holding component **20** is moved in a direction away from the cull block **141** by the link mechanism **223** and the horizontal cam mechanisms **224** and **225**. As a result of this movement, the resin molded product W**2** is moved to the outer side of the protruding portion **141***b* of the cull block **141**.
(81) Moreover, as is shown in FIG. **13**, when the holding component **20** is moved by the horizontal movement mechanism **22**, the holding component **20** is lifted up in conjunction with this movement. More specifically, when the slide component **222***a* is moved in an away direction by the horizontal drive portion **222** (see FIG. **9** (*c*)), the holding component **20** is moved in an upward direction by the vertical cam mechanism **232**. As a result of this movement, the resin molded product W**2** is moved to a height at which it is able to be held by the conveying claws **25**.
(82) Thereafter, as is shown in FIG. **14**, the distance between the conveying claws **25** is shortened, so that the conveying claws **25** are positioned underneath the resin molded product W**2**. In this state, the unloader **19** conveys the resin molded product W**2** that is being held to the storage portion **18** of the storage module **100**C.
Effects of the Present Embodiment
(83) According to the resin molding apparatus **100** of the present embodiment, because the unloader **19** includes the horizontal movement mechanism **22** that moves the holding component **20** which is holding a resin molded product W**2** in a horizontal direction so as to move it away from the cull block **141**, the resin molded product W**2** can be conveyed by the unloader **19** away from the lower mold **15** in which is provided the cull block **141** which includes the protruding portion **141***b* which protrudes above the resin molded product W**2**.
Additional Variant Embodiments
(84) Note that the present invention is not limited to the above-described embodiment.
(85) For example, in the above-described embodiment a structure is employed in which the vertical movement mechanism **23** uses the horizontal drive portion **222** of the horizontal movement mechanism **22**, however, it is also possible to provide a drive portion for the vertical movement mechanism **23** that is different from the horizontal drive portion **222** of the horizontal movement mechanism **22**.
(86) Moreover, in the above-described embodiment, a structure is employed in which the mold release mechanism uses the suction pads **24**, however, it is possible to instead employ a structure in which mold release is achieved by moving the holding component **20** which is holding a resin molded product W**2** in an upward direction, or a structure in which mold release is achieved by moving the conveying apparatus **19** itself in an upward direction. Note that a mold release mechanism is not an indispensable part of the structure, and it is also possible to employ a structure in which the resin molded product W**2** is moved horizontally while in a state of contact with the lower mold **15**.
(87) Furthermore, in the above-described embodiment, a structure is employed in which the holding component **20** is moved in a direction away from the cull block **141** as a result of the slide component **222***a* being moved by the horizontal drive portion **222** in a direction away from the cull bock **141**, however, it is also possible to employ a structure in which the holding component **20** is moved in a direction away from the cull block **141** as a result of the slide component **222***a* being moved by the horizontal drive portion **222** in a direction toward the cull bock **141**. In this case, consideration may be given, for example, to forming the link mechanism **223** using a single link arm (for example, the first link arm **223***a* of the above-described embodiment).
(88) In the above-described embodiment, the conveying claws **25** are used to prevent the resin molded product W**2** which is being held using suction from being dropped, however, it is also possible to employ a structure in which, after a resin molded product W**2** has been lifted up, the holding via suction is released, and the resin molded product W**2** is then held by the conveying claws **25**. Moreover, if a structure is employed in which a resin molded product W**2** is held via suction while being transported, then this structure negates the need for the conveying claws **25** to

be provided.
(89) In the above-described embodiment, a structure is employed in which the cull block **141** is provided in the lower mold **15**, however, it is also possible to employ a structure in which the cull block **141** is provided in the upper mold **16**. In this case, in the same type of operation as is performed in the above-described embodiment, a resin molded product W**2** is transported away from the upper mold **16** by the unloader **19**.
(90) The resin molding apparatus of the present invention is not limited to being used in transfer molding, and may also be used, for example, in compression molding.
(91) While preferred embodiments of the invention have been described and illustrated above, it should be understood that these are exemplary of the invention and are not to be considered as limiting. Additions, omissions, substitutions, and other modifications can be made without departing from the spirit or scope of the present invention. Accordingly, the invention is not to be considered as limited by the foregoing description and is only limited by the scope of the appended claims.
INDUSTRIAL APPLICABILITY
(92) According to the present invention, it is possible for a resin molded product to be conveyed by a conveying apparatus away from one mold in which is provided a resin injection portion which has a component that protrudes between at least a portion of the resin molded product and another mold.

Claims

1. A conveying apparatus that conveys a resin molded product away from one mold in which a resin injection portion is provided, wherein the resin injection portion comprises a component that protrudes between at least a portion of the resin molded product and an other mold, the conveying apparatus comprising: a holding component that holds the resin molded product; a base component in which the holding component is movably provided; a horizontal movement mechanism that moves the holding component which is holding the resin molded product in a horizontal direction so as to move the holding component away from the resin injection portion, wherein the horizontal movement mechanism comprises a guide mechanism that guides the holding component in a direction away from the resin injection portion, and a horizontal drive portion that causes the holding component to move along the guide mechanism, wherein the horizontal drive portion comprises a slide component that slides in a direction towards or in a direction away from the resin injection portion, and wherein the horizontal movement mechanism further comprises a link mechanism that enables the holding component to move in a direction away from the resin injection portion in conjunction with a movement of the slide component, wherein a first horizontal cam mechanism is provided between the slide component and the link mechanism, and a second horizontal cam mechanism is provided between the link mechanism and the holding component; and a vertical movement mechanism that moves the holding component which has been moved by the horizontal movement mechanism in a direction away from the one mold and towards the other mold, wherein the vertical movement mechanism comprises a guide mechanism that guides the holding component in a vertical direction, and a vertical cam mechanism that enables the holding component to move in the vertical direction in conjunction with horizontal movement of the holding component performed by the horizontal drive portion.
2. The conveying apparatus according to claim 1, wherein the conveying apparatus further comprises a mold release mechanism that releases the resin molded product from the one mold, and the horizontal movement mechanism moves the holding component in a horizontal direction in a state in which the resin molded product has been released from the one mold by the mold release mechanism.
3. The conveying apparatus according to claim 2, wherein the holding component comprises

suction pads that suction the resin molded product, and the mold release mechanism is formed using suction force from the suction pads.

4. The conveying apparatus according to claim 1, further comprising conveying claws that hold the resin molded product by grasping it, and after the holding component has been moved in a direction away from the one mold towards the other mold using the vertical movement mechanism, the resin molded product is held using the conveying claws.

5. A resin molding apparatus comprising the conveying apparatus according to claim 1.

6. A conveying method for a resin molded product the employs the conveying apparatus according to claim 1, in which the holding component that is holding a resin molded product which has been resin molded is moved in a horizontal direction by the horizontal movement mechanism so as to move away from the resin injection portion, and the holding component that has been moved away from the resin injection portion is then moved by a vertical movement mechanism in a direction of the other mold.

7. A resin molded product manufacturing method in which an electronic component is resin molded via resin molding, comprising: a molding step in which resin molding is performed on an object to be molded; and a conveying away step in which a resin molded product that has been resin molded is conveyed away using the conveying apparatus according to claim 1.